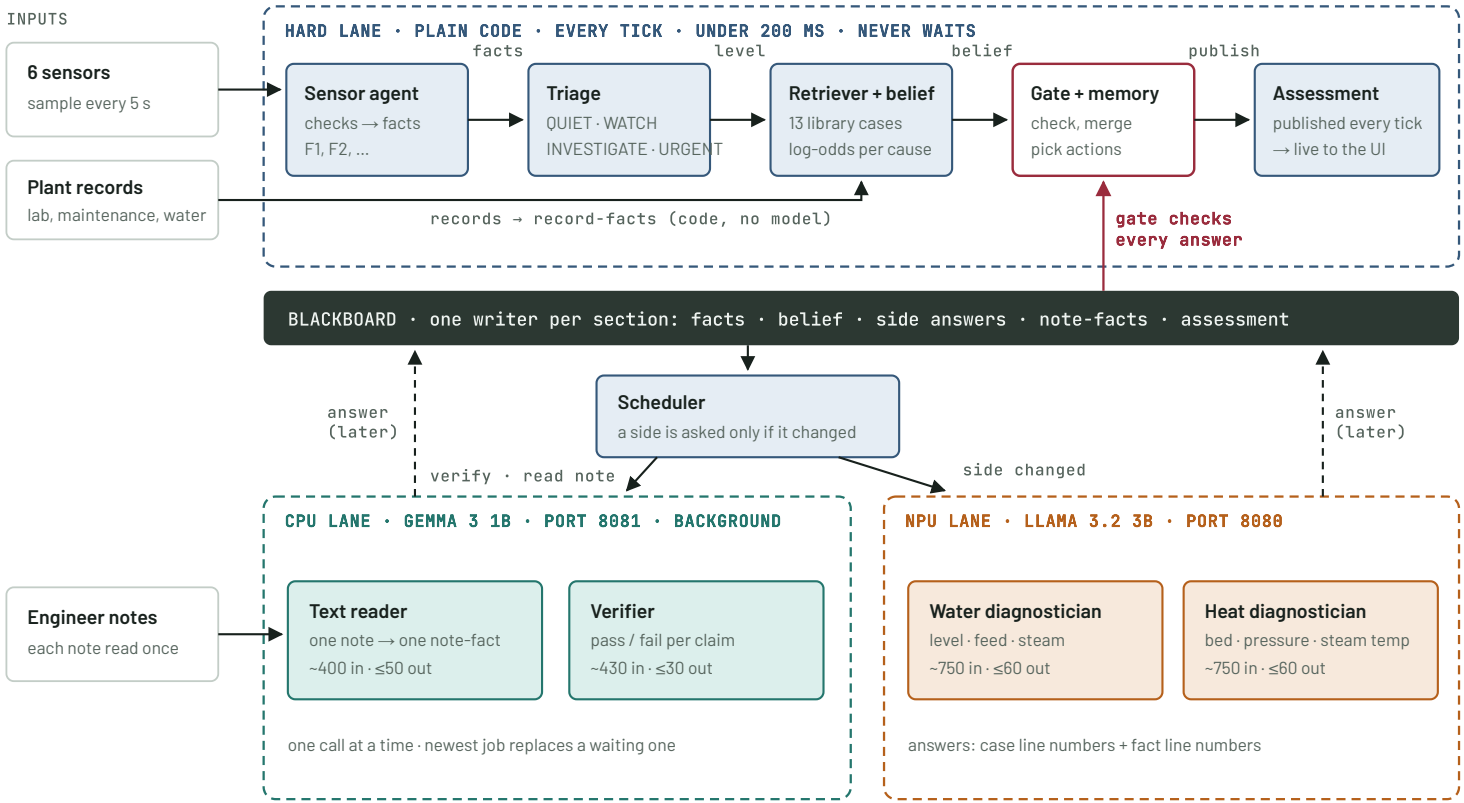
FIELDMIND · 67 TPH AFBC BOILER · QIDK (SNAPDRAGON 8 GEN 3)

How FieldMind works

Every 30 seconds plain code turns six sensor readings into facts and publishes an assessment. Small model calls run in the background on the board's NPU and CPU and refine that assessment when their answers arrive. The model never sees raw numbers and never has the last word.

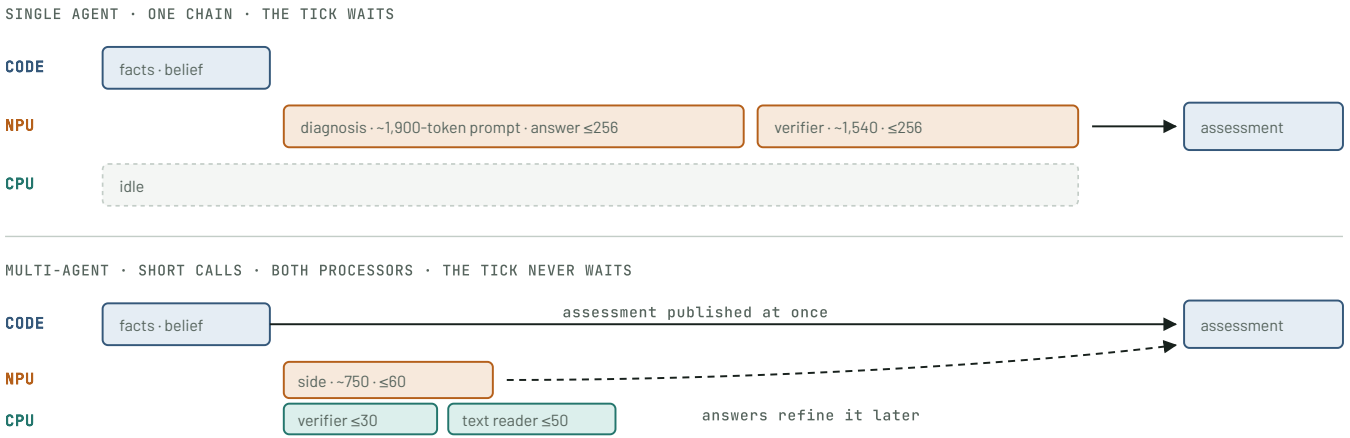
plain code, every tick · NPU lane · Llama 3.2 3B · CPU lane · Gemma 3 1B · checks and drops

One tick, in the demonstration setup



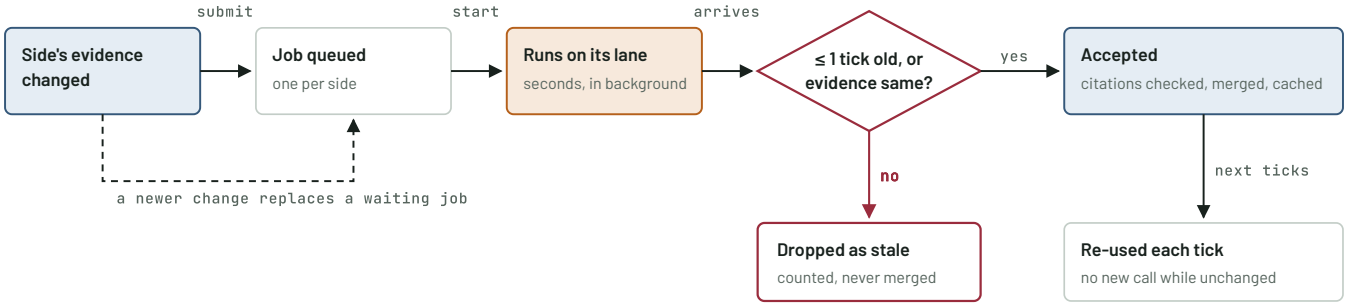
Everything in the top band runs on every tick and publishes an assessment without waiting. Model jobs are sent only when a side's evidence changed. Their answers come back on a later tick, go through the gate (citations must name real facts; the answer must be fresh), and only then change the published ranking.

What changed from the single agent



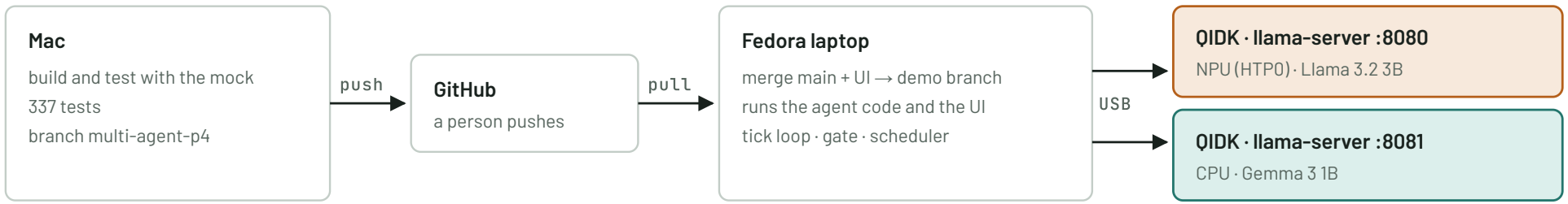
Bars show order and which processor does the work, not measured durations. Prompt sizes are measured token counts on the development episodes (Llama 3.2 tokenizer); answer sizes are the caps. On the board, writing the answer is the slow part (about 16 tokens per second on the NPU), which is why answers were cut from 256 tokens to 60 or less.

The life of a model answer



An answer is used if it arrives within about one tick, or later while its side's evidence has not changed. That is the plan's rule. A lane slower than this would see its answers dropped, so the demonstration tick (30 s) must stay longer than the slowest call.

Where each part runs



Only model calls run on the board today; moving the agent itself onto the board is later Phase 4 work.

The agent's code runs on the laptop and sends each model call over USB (adb port forwarding) to one of two servers on the board. Both servers start once and keep their model loaded.

How far each phase has got

PHASE	WHAT IT ADDS	STATE
0 · Fix the single agent	trustworthy baseline, board setup, model files	done
1 · Plumbing	blackboard, jobs, lanes; same decisions as the single agent	done
2 · Shrink the prompt	prompts ~1,900 → ~750 tokens; line-number answers; text reader	done on mock · one gate count open
3 · Split by plant side	water and heat diagnosticians; call only on change (calls about halved)	done on mock
4 · Real time (demo part)	wall-clock ticks, background lanes, two models, UI hook	done on mock · board next
4 · Rest	drift watcher for slow fouling; agent on the board	not started
5 · Scheduling study	placement policies, energy per correct diagnosis	not started

> Every number on this page comes from the mock backend or from token counts. The mock does no reasoning, so nothing here is a model accuracy result. Board numbers come from the demonstration run.